

Scientific Reasoning

HLE (Humanity's Last Exam)

- Description: Recent frontier academic benchmark from the Centre for AI Safety (led by Dan Hendrycks).
- Paper: <https://arxiv.org/abs/2501.14249v2>
- Dataset: <https://huggingface.co/datasets/cais/hle>
- Implementation:
 - 2,158 text-only questions across mathematics, humanities and the natural sciences (from the May 2025 revision which contains 2,500 total questions — we use the text-only subset for maximum comparability across models)
 - We note that the HLE authors disclose that their dataset curation process involved adversarial selection of questions based on tests with GPT-4o, Gemini 1.5 Pro, Claude 3.5 Sonnet, o1, o1-mini, and o1-preview (latter two for text-only questions only). We therefore discourage direct comparison of these models with models that were not used in the HLE curation process, as the dataset is potentially biased against the models used in the curation process.
 - Evaluated with an equality checker LLM prompt adapted from the original HLE paper, using GPT-5.6 Luna (medium), with pass@1 scoring (find prompt below)

CritPt

- Description: Research-level physics reasoning benchmark with unpublished, frontier physics problems spanning a wide range of subfields.
- Paper: <https://arxiv.org/abs/2509.26574>
- Website: <https://critpt.com/>
- Repository: <https://github.com/CritPt-Benchmark/CritPt>
- Dataset: <https://huggingface.co/datasets/CritPt-Benchmark/CritPt>
- Implementation:
 - We implement the 'challenge' level components for all 70 test-set challenges (the example challenge is excluded) in collaboration with the CritPt team
 - We run 5 repeats for each question with pass@1 scoring
 - The models are called with a two-step parsing approach, where the first step requests that the model complete the challenge with reasoning, and the second step formats the response into the expected code format for grading (see example prompt for parsing on the CritPt evaluation page)
 - Token usage and cost estimates reflect both steps (reasoning and answer parsing)
 - Answer formats include numerical values, symbolic expressions in SymPy, and Python functions (evaluated with test cases)
 - The official CritPt grading server is used to assess all challenge responses for correctness. Grading API access is granted case by case to approved labs and researchers — email critpt@artificialanalysis.ai to request it, and see the [Artificial Analysis API documentation](#) for details